

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – VI

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Class: TYIT	Batch: 1
Date of Assignment: 30/07/2026	Date/Time of Submission: 30/07/2026

Part B - Program

- a. Create a simple website for a College Information Portal and convert it into a Progressive Web Application.

Code:

Manifest.json

```
{
  "name": "HTML TO PWA",
  "short_name": "HTP",
  "start_url": "/",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "/icons/icon1.png",
      "sizes": "128x128",
      "type": "image/png",
      "purpose": "any"
    },
    {
      "src": "/icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png",
      "purpose": "any"
    }
  ]
}
```

service-worker.js

```
self.addEventListener("install", e => {
```

```
    console.log("Service Worker installed");
  });
```

```
self.addEventListener("fetch", e => {
  event.respondWith(
    fetch(event.request)
  );
});
```

_Layout.cshtml

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="utf-8" />

<meta name="viewport" content="width=device-width, initial-scale=1.0" />

<title>@ViewData["Title"] - HTML_to_PWA</title>

<script type="importmap"></script>

<link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />

<link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />

<link rel="stylesheet" href="~/HTML_to_PWA.styles.css" asp-append-version="true" />

<link rel="manifest" href="~/manifest.json" />

</head>

<body>

<header>

<nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-white border-bottom box-shadow mb-3">

<div class="container-fluid">

HTML_to_PWA

<button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target=".navbar-collapse" aria-controls="navbarSupportedContent" aria-expanded="false" aria-label="Toggle navigation">

</button>

<div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">

<ul class="navbar-nav flex-grow-1">

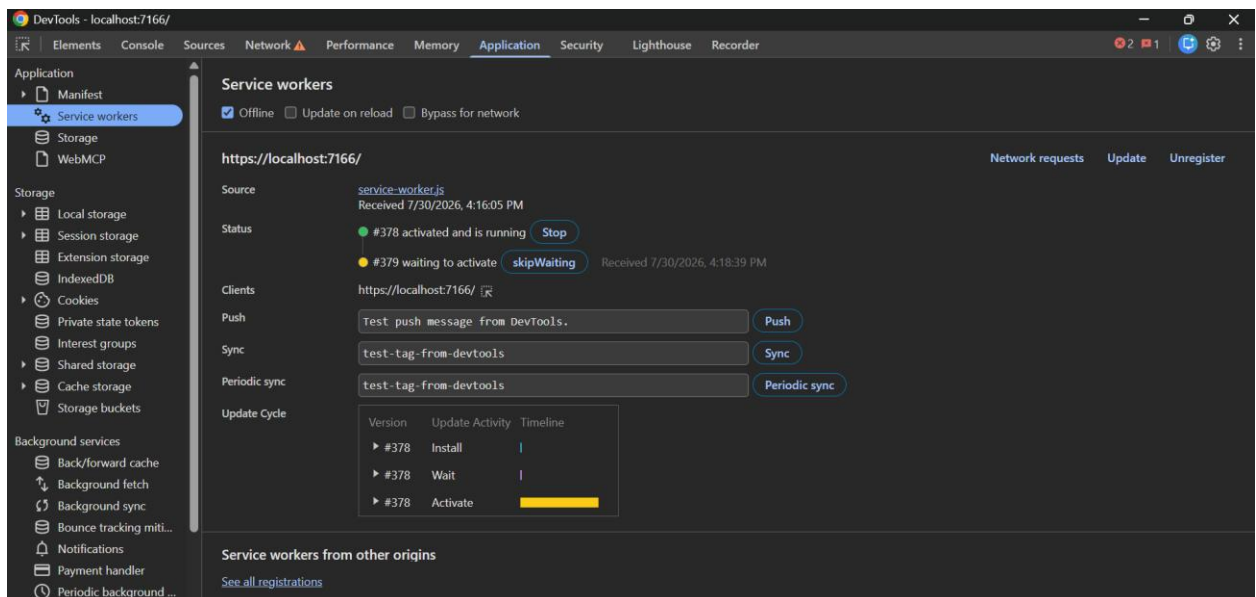
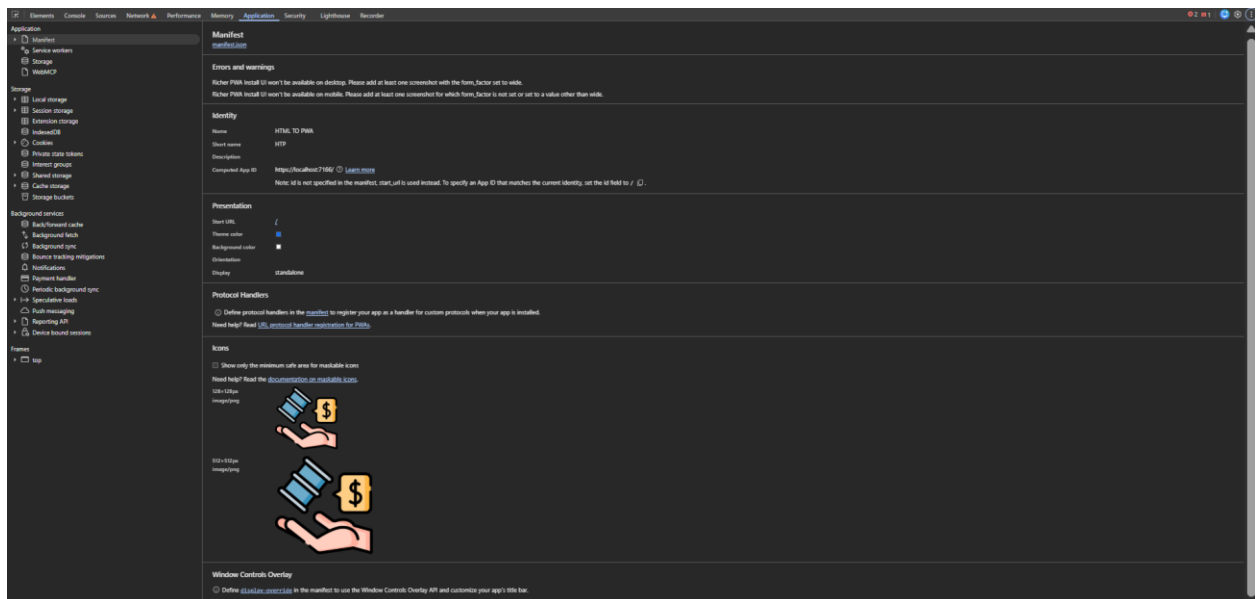
<li class="nav-item">

Home

```
        </li>
        <li class="nav-item">
            <a class="nav-link text-dark" asp-area="" asp-controller="Home" asp-
action="Privacy">Privacy</a>
        </li>
    </ul>
</div>
</div>
</nav>
</header>
<div class="container">
    <main role="main" class="pb-3">
        @RenderBody()
    </main>
</div>

<footer class="border-top footer text-muted">
    <div class="container">
        &copy; 2026 - HTML_to_PWA - <a asp-area="" asp-controller="Home" asp-
action="Privacy">Privacy</a>
    </div>
</footer>
<script src="~/lib/jquery/dist/jquery.min.js"></script>
<script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
<script src="~/js/site.js" asp-append-version="true"></script>
@await RenderSectionAsync("Scripts", required: false)
<script>
if ('serviceWorker' in navigator) {
    navigator.serviceWorker.register('/service-worker.js')
        .then(() => console.log('Service Worker Registered'))
        .catch(error => console.error('Registration failed:', error));
}
</script>
</body>
</html>
```

Output:



- b. Create a PWA that displays a custom offline page when the user is not connected to the internet.

Code:

Manifest.json

```
{
  "name": "HTML TO PWA",
  "short_name": "HTP",
  "start_url": "/",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
```

```
"icons": [  
  {  
    "src": "/icons/icon1.png",  
    "sizes": "128x128",  
    "type": "image/png",  
    "purpose": "any"  
  },  
  {  
    "src": "/icons/icon2.png",  
    "sizes": "512x512",  
    "type": "image/png",  
    "purpose": "any"  
  }  
]  
}  
service-worker.js  
const CACHE = "v1";  
  
self.addEventListener("install", e => {  
  e.waitUntil(  
    caches.open(CACHE).then(cache => {  
      return cache.addAll([  
        "/",  
        "/offline.html"  
      ]);  
    })  
  );  
  
  console.log("Service Worker Installed");  
});  
  
self.addEventListener("fetch", e => {  
  e.respondWith(  
    fetch(e.request).catch(() => caches.match("/offline.html"))  
  );  
});  
  
_Layout.cshtml  
<!DOCTYPE html>  
<html lang="en">  
<head>
```

```
<meta charset="utf-8" />
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
<title>@ViewData["Title"] - HTML_to_PWA</title>
<script type="importmap"></script>
<link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
<link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
<link rel="stylesheet" href="~/HTML_to_PWA.styles.css" asp-append-
version="true" />
<link rel="manifest" href="~/manifest.json" />
</head>
<body>
  <header>
    <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-
white border-bottom box-shadow mb-3">
      <div class="container-fluid">
        <a class="navbar-brand" asp-area="" asp-controller="Home" asp-
action="Index">HTML_to_PWA</a>
        <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
data-bs-target=".navbar-collapse" aria-controls="navbarSupportedContent"
aria-expanded="false" aria-label="Toggle navigation">
          <span class="navbar-toggler-icon"></span>
        </button>
        <div class="navbar-collapse collapse d-sm-inline-flex justify-content-
between">
          <ul class="navbar-nav flex-grow-1">
            <li class="nav-item">
              <a class="nav-link text-dark" asp-area="" asp-controller="Home" asp-
action="Index">Home</a>
            </li>
            <li class="nav-item">
              <a class="nav-link text-dark" asp-area="" asp-controller="Home" asp-
action="Privacy">Privacy</a>
            </li>
          </ul>
        </div>
      </div>
    </nav>
  </header>
  <div class="container">
    <main role="main" class="pb-3">
      @RenderBody()
```

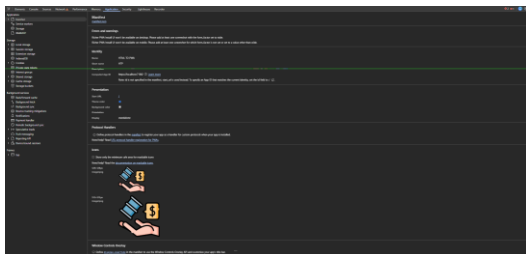
```

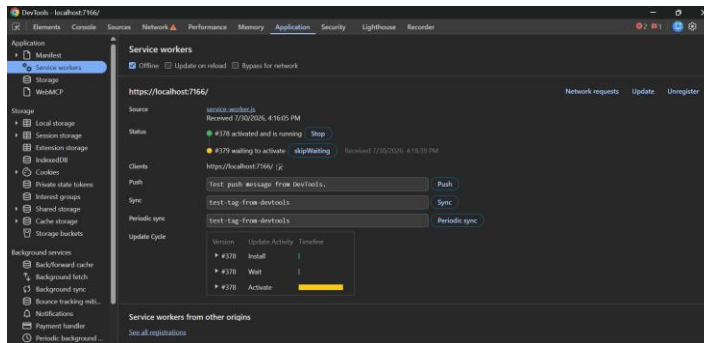
    </main>
  </div>

  <footer class="border-top footer text-muted">
    <div class="container">
      &copy; 2026 - HTML_to_PWA - <a asp-area="" asp-controller="Home" asp-
action="Privacy">Privacy</a>
    </div>
  </footer>
  <script src="~/lib/jquery/dist/jquery.min.js"></script>
  <script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
  <script src="~/js/site.js" asp-append-version="true"></script>
  @await RenderSectionAsync("Scripts", required: false)
  <script>
    if ('serviceWorker' in navigator) {
      navigator.serviceWorker.register('/service-worker.js')
        .then(() => console.log('Service Worker Registered'))
        .catch(error => console.error('Registration failed:', error));
    }
  </script>
</body>
</html>
offline.html
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8" />
  <title>Welcome to college </title>
</head>
<body>
  <h1> No Network </h1>
</body>
</html>

```

Output:





offline